Answer **all** *the questions.*

Mathematical tables may be used in **any** *question. The use of non-programmable, silent and cordless calculator is allowed.*

Each question is followed by **four** *options lettered* A *to* D. *Find the* **correct** *option for* **each** *question and shade in* **pencil**, *on your answer sheet, the answer space which bears the same letter as the option you have chosen.*

Give only **one** *answer to* **each** *question. An example is given below,*

The ages, in years, of four boys are 10, 12, 14 and 18. What is the average age of the boys?

A. 12 years
B. 13½ years
C. 13 years
D. 12½ years

The correct answer is 13½ years, *which is lettered* B, *and therefore answer space* B *would be shaded.*

⊂A⊃ ⊂B⊃ ⊂C⊃ ⊂D⊃

Think carefully before you shade the answer spaces; erase completely any answers you wish to change.

Do all rough work on this question paper.

Now answer the following questions.

1. Mr. Gyan bought a car for $16,800.00 and later sold it at 85 % of the original cost. He spent $3,930.75 from the money he received from the sale of the car and invested the remaining at 18 % per annum simple interest. How much interest did he earn in 9 years?

A. $67,657.00
B. $32,018.32
C. $16,765.79
D. $20,848.19

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2. The locus, L is such that $|PM| = |PN|$. Which of the following best describes L?

A. It is the locus of points perpendicular to the line MN.
B. It is the locus of points parallel to line MN.
C. It is the circle centre P, radius MN.
D. It is the locus of points equidistant from M and N.

3. A town P is due south of town Q and town R is on a bearing of 125° from Q. If town R is 20 km due east of P, find $|PQ|$.

A. 20 km
B. 15 km
C. 12 km
D. 14 km

4. If $V = \pi r^2 h$ and $S = 2\pi rh$, express V in terms of r and S.

A. $V = \frac{2r}{S}$

B. $V = \frac{1}{2}rS$

C. $V = 2rS$

D. $V = \frac{r}{2S}$

5. The mean and range of 1.20, 1.00, 0.90, 1.40, 0.80, 0.80, 1.20 and 1.10 are m and r respectively. Find the value of $(m + r)$

A. 2.55
B. 2.45
C. 1.65
D. 1.85

6. The inverse of the sum of -3 and twice a certain number is $\frac{1}{5}$. Find the number.

A. 1
B. $1\frac{1}{2}$
C. 4
D. $1\frac{3}{5}$

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7. Solve: $\frac{\log_3(2x-1)}{\log_3 243} = \frac{2}{5}$.

A. $x = 3$
B. $x = 4$
C. $x = 6$
D. $x = 5$

8. The first 4 terms of an Arithmetic Progression (A.P) are 8, x, y and 17. Find the value of $(x + y)$.

A. 20
B 25
C. 40
D. 31

9. A man cycles a distance of $(3a)$ km at V km/h and then walks a distance of a km at $(V-7)$ km/h. Find the total number of hours he spent travelling.

A. $\frac{3a}{V}+\frac{a}{V-7}$

B. $\frac{V}{3a}+\frac{V-7}{a}$

C. $\frac{2V-7}{4a}$

D. $\frac{4a}{2V-7}$

10. A function is defined by $g: x \rightarrow \frac{2x-1}{3x-2}$. For what value of x is g not defined?

A. $\frac{2}{3}$

B. $-\frac{2}{3}$

C. $-\frac{1}{2}$

D. $\frac{1}{2}$

11. A box contains 12 red, 8 yellow and 10 green pebbles, all of the same size. If two pebbles are selected at random one after the other, **with replacement**, find the probability that the first is green and the second is not a red pebble.

A. $\frac{1}{5}$

B. $\frac{1}{9}$

C. $\frac{17}{20}$

D. $\frac{15}{20}$

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12. The sets $P = \{x: x = 3, 5, 6\}$ and $Q = \{x: 2 \leq x \leq 5\}$ are subsets of $\mu = \{x: 1 \leq x \leq 10$, where x is an integer$\}$. Find $P' \cap Q$.

A. $\{1, 4\}$

B. $\{2, 4\}$

C. $\{3, 4\}$

D. $\{4, 5\}$

13. Given that $\log_{10}5 = x$ and $\log_{10}11 = y$, express $\log_{10}275$ in terms of x and y.

A. $2(x+y)$

B. $2x+y$

C. $2xy$

D. $x+2y$

14. Find the coefficient of x^5 in the expansion of $(4x + 5)(3 - 2x)$.
A. 15
B. 8
C. −8
D. 2

15. Find the mean deviation of 2, 5, 7, 9 and 15.
A. 2.45
B. 3.52
C. 5.23
D. 4.14

16. A train is moving at 60 miles per hour. If it passes a sign post in 15 seconds, find, in yards, the length of the train. [Take 1 mile = 1760 yards]
A. 440 yards
B. 420 yards
C. 360 yards
D. 400 yards

17. A school has a total population of 1,800 students. Out of this number $\frac{4}{9}$ are in SHS 1, $\frac{13}{36}$ are in SHS 2 and the rest are in SHS 3. Calculate the number of students in SHS 3.
A. 800
B. 650
C. 350
D. 500

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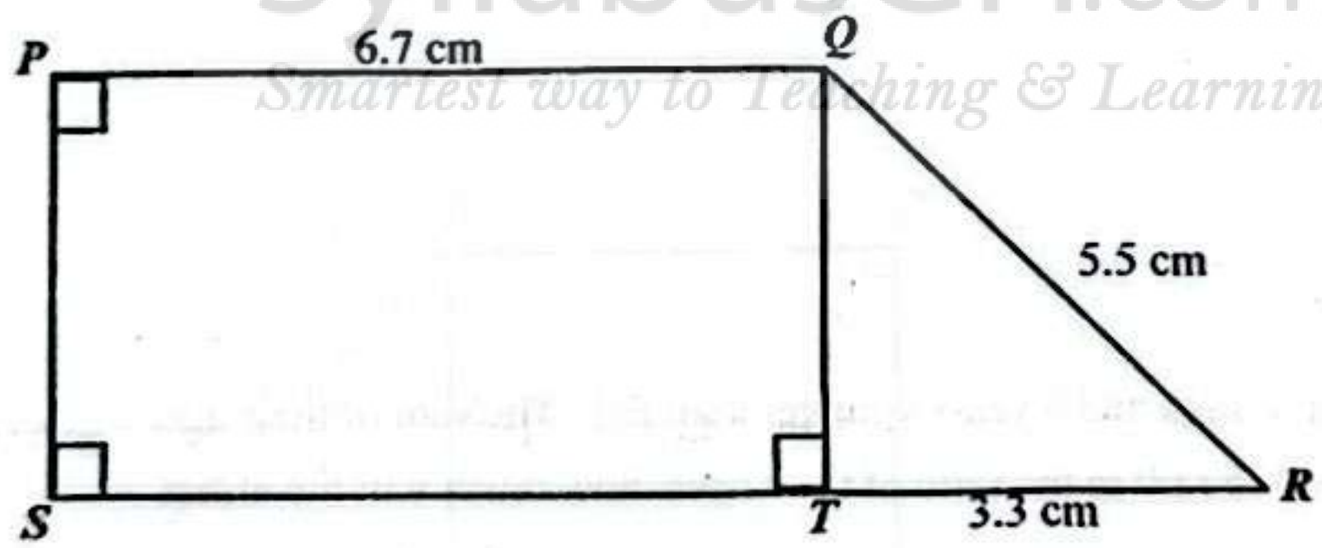
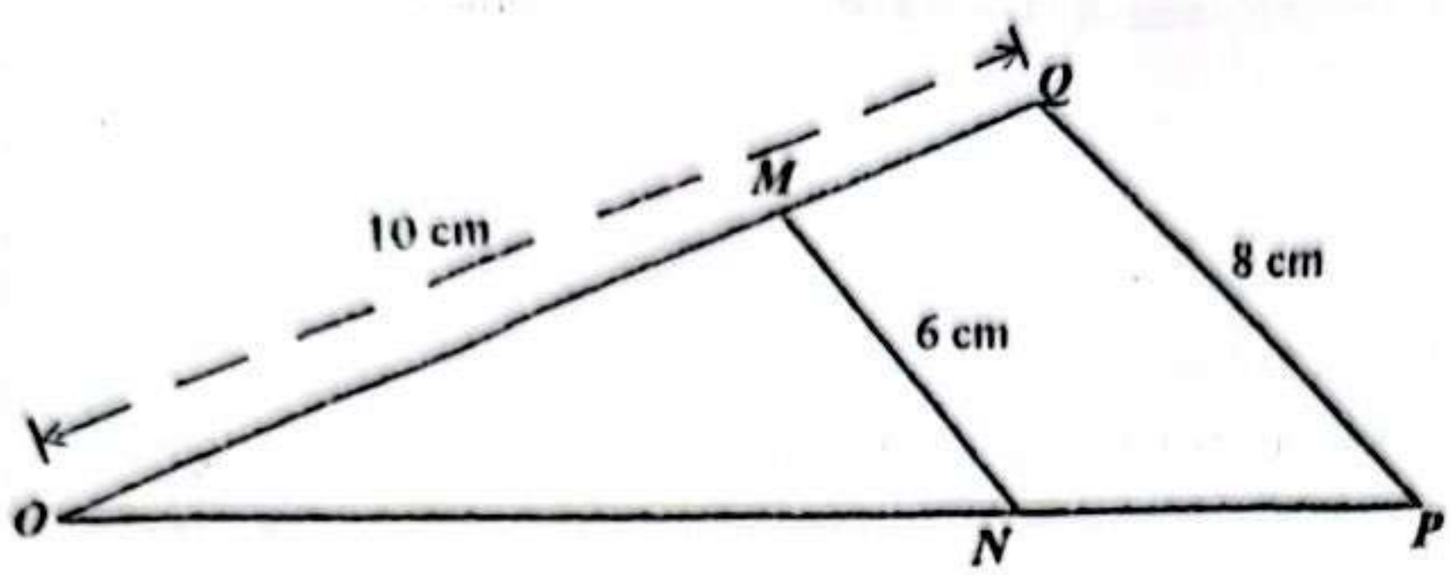
18.



NOT DRAWN TO SCALE

In the diagram, $|PQ| = 6.7$ cm, $|QR| = 5.5$ cm and $|RT| = 3.3$ cm. Find the area.
A. 45.93 cm^2
B. 36.74 cm^2
C. 26.40 cm^2
D. 27.56 cm^2

19.



NOT DRAWN TO SCALE

In the diagram, ΔOMN is similar to ΔOPQ. Find $|MQ|$.

A. 4.0 cm
B. 3.5 cm
C. 2.5 cm
D. 3.0 cm

20. A car consumes 36 gallons of fuel over a distance of 2,268 km. Find, in km per litre, the fuel consumption. [Take 1 gallon = 4.5 litres]

A. 14
B. 15
C. 63
D. 50

21. Given that $\frac{3}{x+y} - \frac{4}{x-y} = 0$, find the value of $\frac{x}{y}$, where $y \neq 0$.

A. 7
B. −7
C. $\frac{1}{7}$
D. $-\frac{1}{7}$

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22. Kwaku is 2 years older than Atanga and 6 years younger than Esi. The sum of their ages is 70. If they decide to share \$153.00 in the ratio of their ages, how much will the eldest person receive?

A. \$61.20
B. \$51.00
C. \$43.71
D. \$48.09

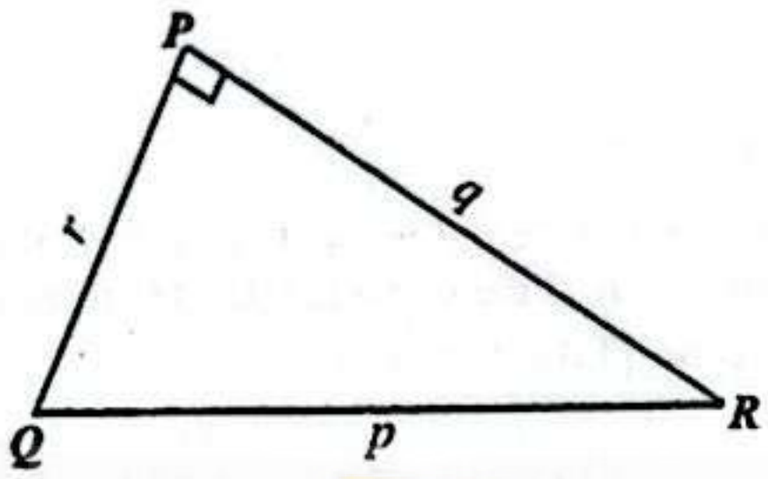
23. A scientific sensor records air pollution levels in base 3. If the highest recorded pollution level for a week was 2120_3, what is the decimal equivalent?

A. 62
B. 65
C. 72
D. 69

24. The straight line $3y + bx - 6 = 0$ passes through (3, 4). Find the value of b.

A. 12
B. 8
C. −8
D. −2

25.



NOT DRAWN TO SCALE

In the diagram $\angle QPR = 90^\circ$. If $q^2 = 25 - r^2$, find the value of p.

A. 4
B. 5
C. 6
D. 3

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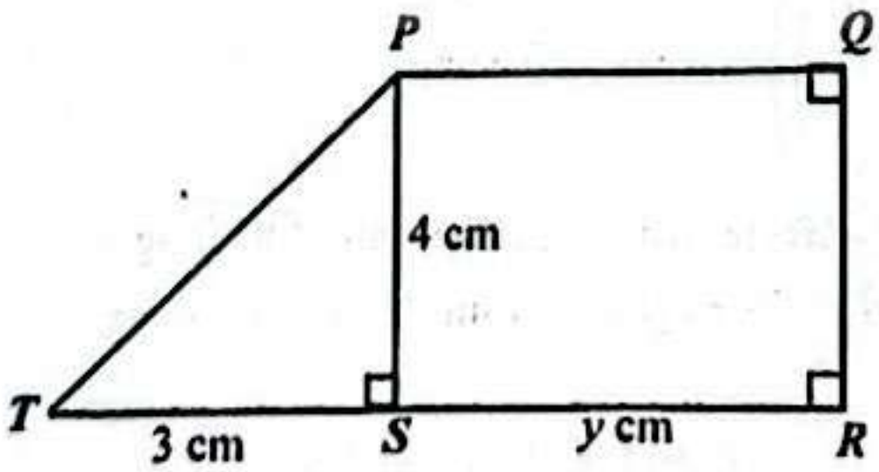
26.



NOT DRAWN TO SCALE

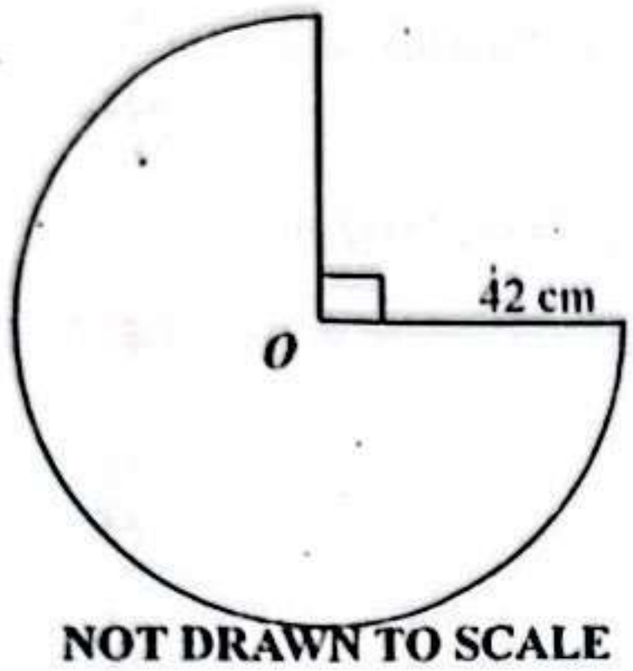
In the diagram $PQRST$ is a trapezium. $|ST| = 3$ cm, $|RS| = y$ cm and $|PS| = 4$ cm.
For what values of y will the area of the trapezium $PQRST$ be less than or equal to 10 cm^2?

A. $y \leq 4$ cm
B. $y \leq 1$ cm
C. $y \leq 6$ cm
D. $y \leq 2$ cm

27. Correct, 5.40514 to **three** significant figures.

A. 541
B. 5.41
C. 5.40
D. 5.45

28.

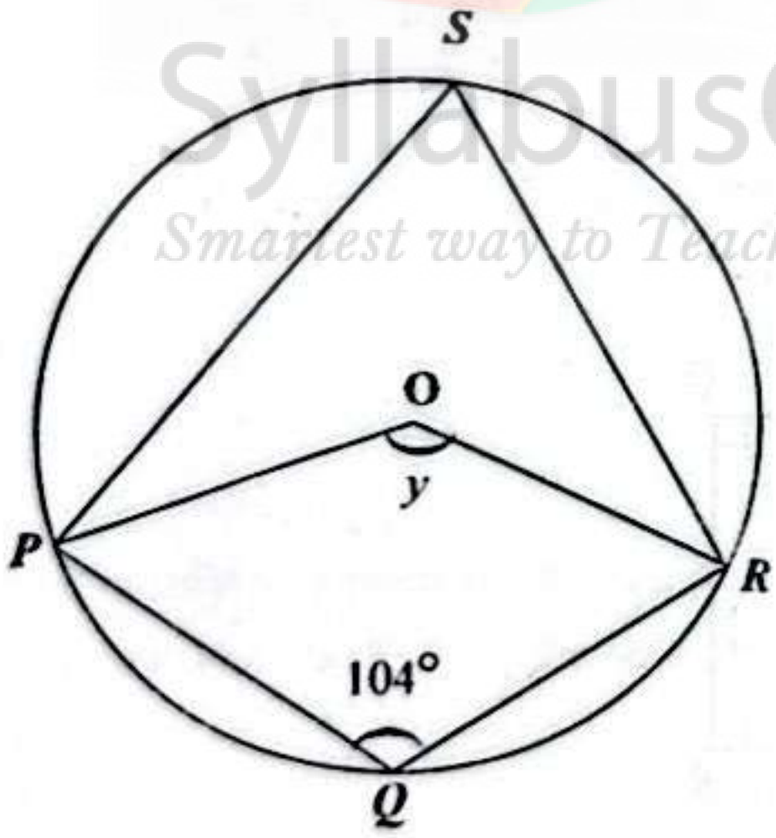


NOT DRAWN TO SCALE

The diagram shows a sector of a circle of radius 42 cm. If the sector is folded to form a right circular cone, find the surface area of the cone. [Take $\pi = \frac{22}{7}$]

A. 4,158 cm^2
B. 5,544 cm^2
C. 29,106 cm^2
D. 11,088 cm^2

29.



NOT DRAWN TO SCALE

In the diagram, P, Q, R and S are points on the circle centre O and $\angle PQR = 104°$. Find the angle marked y.

A. 152°
B. 76°
C. 208°
D. 52°

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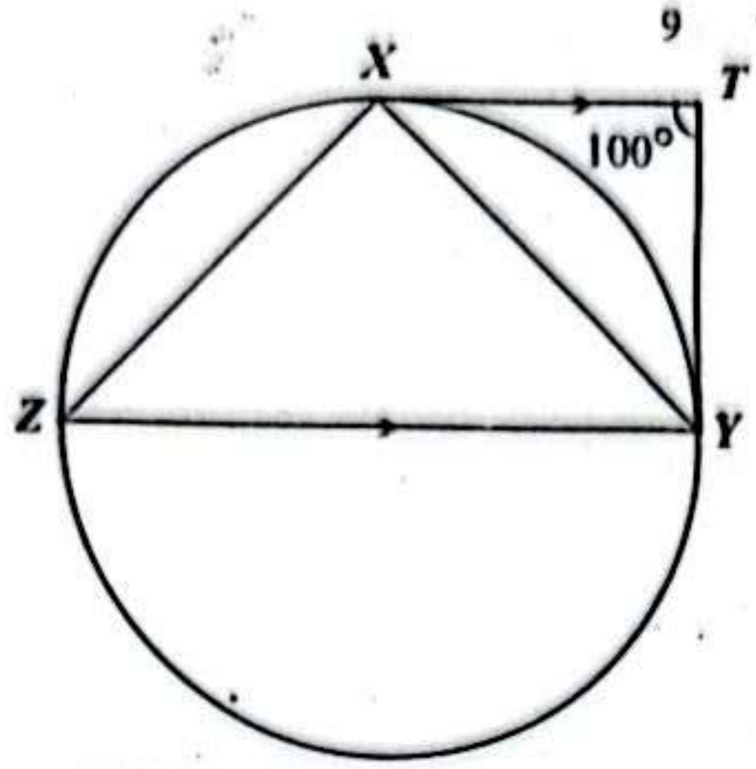
30.



NOT DRAWN TO SCALE

In the diagram $\overline{TX}$ and $\overline{TY}$ are tangents to the circle XYZ at X and Y respectively. $\overline{TX} \parallel \overline{YZ}$ and $\angle XTY = 100°$. Find $\angle ZXY$.

A. 130°
B. 100°
C. 80°
D. 90°

31. Given that $\left(1\frac{2}{3}\right)^{\frac{x}{2}} = \left(4\frac{17}{27}\right)$, find the value of x.

A. 2
B. 3
C. 6
D. 5

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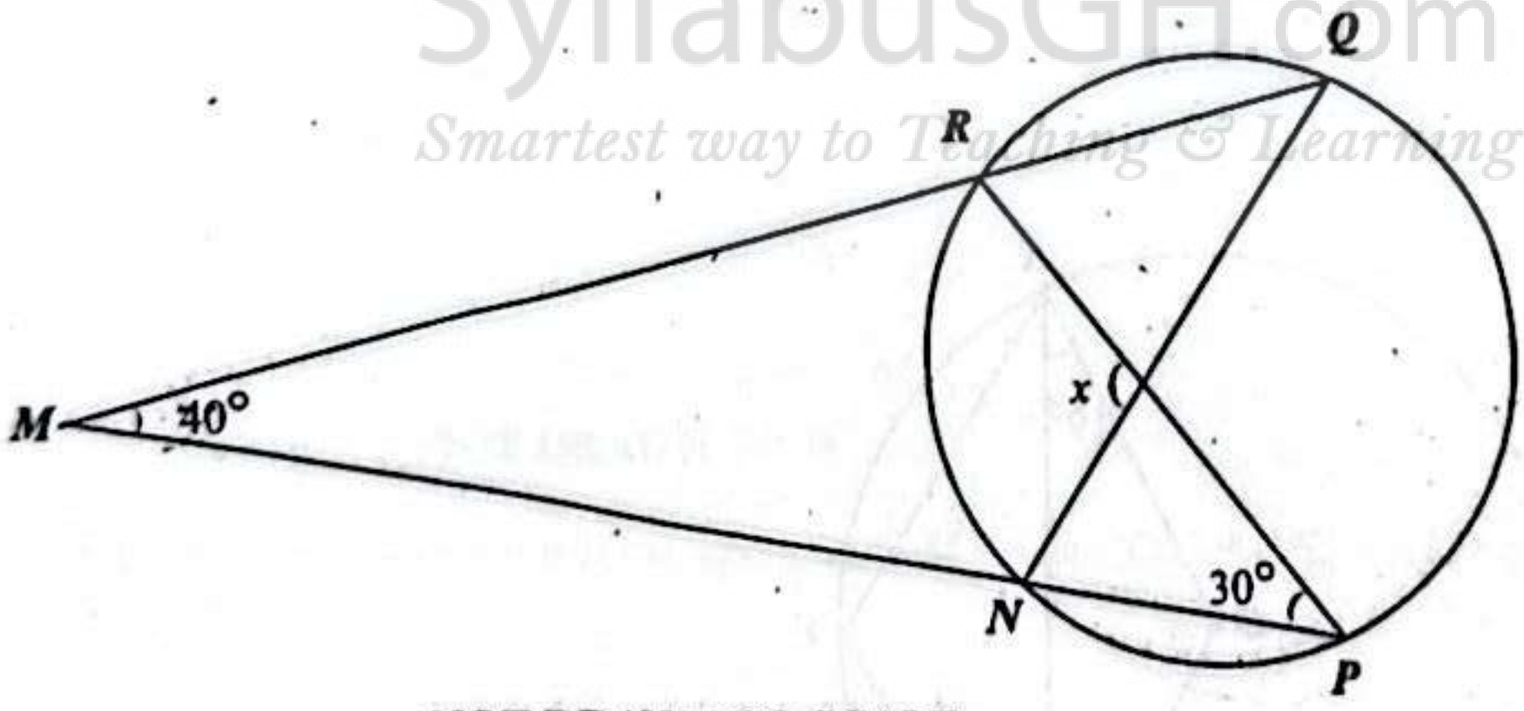
32.



NOT DRAWN TO SCALE

In the diagram, $\angle QMP = 40°$ and $\angle RPM = 30°$. Find the value of x.

A. 40°
B. 70°
C. 110°
D. 100°

33. A fair dice is thrown once. What is the probability of obtaining a 3 or 5?

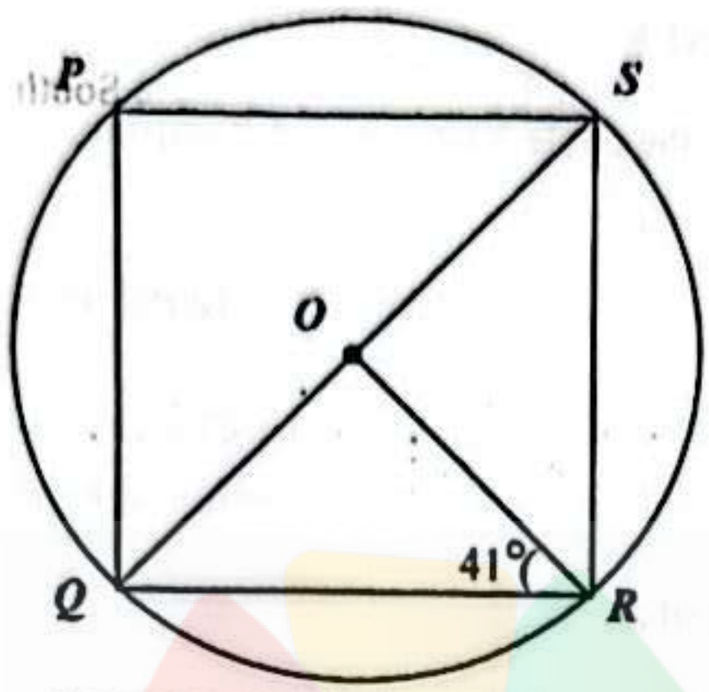
A. $\frac{1}{6}$

B. $\frac{2}{3}$

C. $\frac{1}{2}$

D. $\frac{1}{3}$

34.



NOT DRAWN TO SCALE

In the diagram, P, Q, R and S are points on the circle centre O. $\overline{QS}$ is a diameter and $\angle ORQ = 41°$. Find $\angle QSR$.

A. 41°

B. 49°

C. 82°

D. 60°

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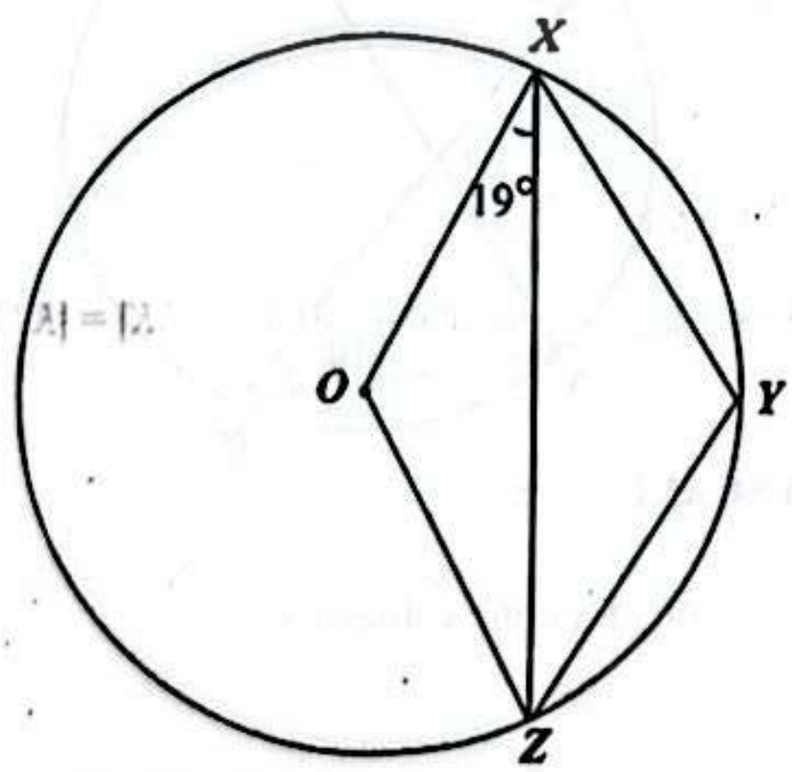
35.



NOT DRAWN TO SCALE

In the diagram, O is the centre of the circle and $\angle OXZ = 19°$. Find $\angle XYZ$.

A. 142°

B. 109°

C. 68°

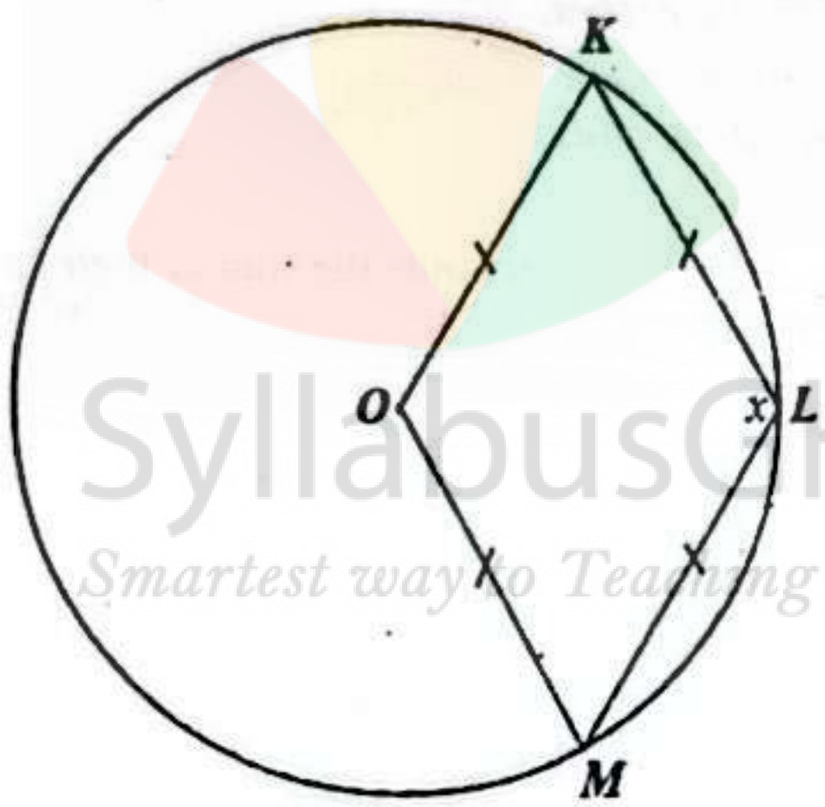
D. 71°

36. The mean of a, b, c, d and e is 15. Calculate the mean of $(a + 1)$, $(b + 3)$, $(c + 5)$, $(d + 7)$ and $(e + 9)$.
A. 40
B. 25
C. 15
D. 20

37. The last term of the sequence: 7, 11, 15, ... is 115. Find the number of terms in the sequence.
A. 29
B. 28
C. 26
D. 27

38. Peter is in the East of Tom and Tom is in the North of John. If Mike is in the South of John, in which direction of Peter is Mike?
A. South East
B. South West
C. North East
D. South

39.



NOT DRAWN TO SCALE

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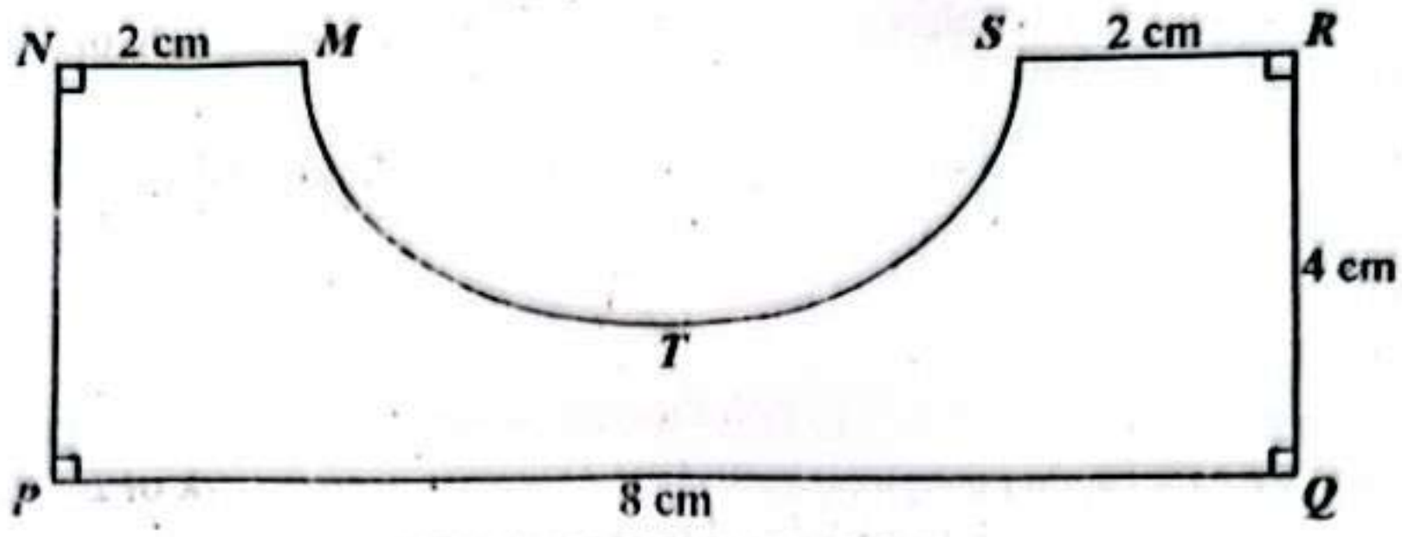
In the diagram, O is the centre of the circle, $\angle KLM = x$ and $|OM| = |ML| = |LK| = |KO|$. Find $\angle OKL$.
A. 15°
B. 30°
C. 120°
D. 60°

40.



NOT DRAWN TO SCALE

The diagram shows a field *MNPQRST*, where *MTS* is a semicircle. Find the area of the field. [Take $\pi = \frac{22}{7}$]

A. $25.7\ cm^2$
B. $26.3\ cm^2$
C. $28.6\ cm^2$
D. $27.5\ cm^2$

41. The radius of a circle is 4 times the radius of another circle. What is the relationship between their areas?

A. The area of one is 64 times that of the other.
B. The area of one is 16 times that of the other.
C. The area of one is 8 times that of the other.
D. The area of one is 12 times that of the other.

42. Gifty is 19 years younger than her mother. In three years time the sum of their ages will be 67 years. Find the sum of their ages now.

A. 22 years
B. 40 years
C. 61 years
D. 48 years

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43. Factorize: $5p^2 + 4pq - 15pr - 12qr$.

A. $(p - 3r)(5p + 4q)$
B. $(p + 3r)(5p - 4q)$
C. $(p - 3r)(5p - 4q)$
D. $(p + 3r)(5p + 4q)$

44. Some of the interior angles of a pentagon are 111°, 131° and 79°. The remaining angles are x and y in the ratio 2 : 3 respectively. Find, correct to the nearest whole number, the value of x.

A. 44°
B. 56°
C. 88°
D. 80°

45. A pyramid has a rectangular base of length 6.5 cm and width 4.2 cm. If the height of the pyramid is 12 cm, find the volume.

A. $109.2\ cm^3$

B. $32.70\ cm^3$

C. $327.6\ cm^3$

D. $10.20\ cm^3$

46. In a test, the mean mark of 25 students in a class, P, is 72.2 and the mean mark of 20 students in another class, Q, is 65. Find the mean mark of all the students in P and Q.

A. 70

B. 69

C. 66

D. 68

47. The sum of the ages of Esi and Amina is 24 years and the difference in their ages is 6 years. If Amina is older than Esi, find Esi's age.

A. 9

B. 12

C. 15

D. 13

48. Solve: $4x + 1 = 3 (\text{mod } 11)$.

A. 3

B. 4

C. 6

D. 5

49. Simplify: $\frac{25p^2 - q^2}{5p^2 + 6pq + q^2}$.

A. $\frac{5p - q}{p - q}$

B. $\frac{5p - q}{p + q}$

C. $\frac{5p + q}{p - q}$

D. $\frac{5p + q}{p + q}$

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50. 21. Make w the subject of the relation $\frac{1}{w} = (h - 1)\left(\frac{1}{m} - \frac{1}{n}\right)$.

A. $w = \frac{mn}{(h - 1)(n - m)}$

B. $w = \frac{mn}{(h - n)(1 - m)}$

C. $w = \frac{mn}{(m - 1)(h - n)}$

D. $w = \frac{mn}{(n + 1)(h - m)}$

END OF PAPER